

Lesson 1

Overview

The term AI was introduced by Prof. John McCarthy at a conference at Dartmouth College in 1956. McCarthy defines AI as the “**science and engineering of making intelligent machines, especially intelligent computer programs**”.

You interact with AI systems daily but might not be aware of it. Every time that you use a search engine such as Google or Bing, explore news websites such as the BBC or the New York Times, talk to a virtual assistant such as Siri, or use an automated language translation service, you are dealing with intelligent systems. Generally, AI occupies a wide landscape and there are many potential uses for it.

The objective of this course is to familiarize you with AI, which increases its influence over our daily lives. The course provides the information and tools that you need to start your journey into the cognitive or AI era of computing. You start the journey by defining AI.

What artificial intelligence is?

AI is a sub field of computer science that aims at building computer systems that can perform tasks that normally require human intelligence.

For years, the challenging goal of AI has been developing computer systems that equal or exceed human intelligence. AI-based machines are intended to perceive their environment and take actions that optimize their level of success.

AI research uses techniques from many fields, such as linguistics, economics, and psychology. These techniques are used in applications, such as control systems, natural language processing, facial recognition, speech recognition, business analytics, pattern matching, and data mining.

Since the invention of computers or machines, their capability to perform various tasks went on growing exponentially. Humans have developed the power of computer systems in terms of their diverse working domains, their increasing speed, and reducing size with respect to time.

A branch of Computer Science named *Artificial Intelligence* pursues creating the computers or machines as intelligent as human beings.

What is Artificial Intelligence?

According to the father of Artificial Intelligence, John McCarthy, it is “*The science and engineering of making intelligent machines, especially intelligent computer programs*”.

Artificial Intelligence is a way of **making a computer, a computer-controlled robot, or a software think intelligently**, in the similar manner the intelligent humans think.

AI is accomplished by studying how human brain thinks, and how humans learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems.

Artificial Intelligence is a branch of *Science* which deals with helping machines find solutions to complex problems in a more human-like fashion. This generally involves borrowing characteristics from human intelligence, and applying them as algorithms in a computer friendly way. A more or less flexible or efficient approach can be taken depending on the requirements established, which influences how artificial the intelligent behavior appears.

Philosophy of AI

While exploiting the power of the computer systems, the curiosity of human, lead him to wonder, “*Can a machine think and behave like humans do?*”

Thus, the development of AI started with the intention of creating similar intelligence in machines that we find and regard high in humans. Unlike humans, computers have trouble understanding specific situations, and adapting to new situations. Artificial Intelligence aims to improve machine behaviour in tackling such complex tasks.

Goals of AI

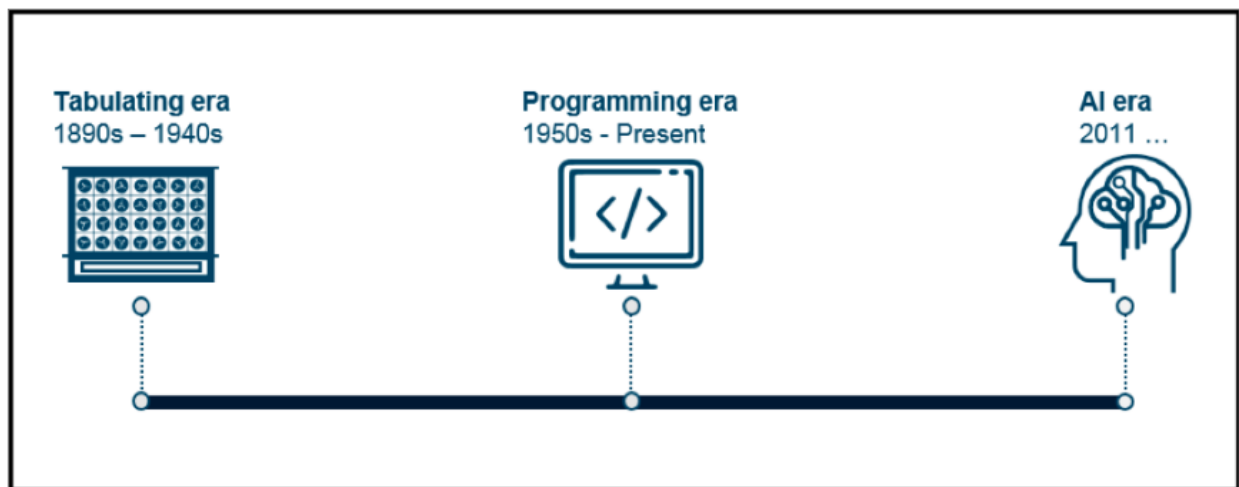
To Create Expert Systems –The systems which exhibit intelligent behavior, learn, demonstrate, explain, and advice its users.

To Implement Human Intelligence in Machines – Creating systems that understand, think, learn, and behave like humans.

THE THREE ERAS OF COMPUTING

To understand the future of AI, placing it in the historical context is important. To date, two distinct eras of computing have occurred: the tabulating era and the programming era. We are entering the third and most transformational era in computing's evolution, the AI computing era (also known as the cognitive era).

Figure 1 shows the timeline for the three eras of computing.



The three eras are described in the following sections.

Tabulating era (1890s – 1940s)

The first era of computing consisted of single-purpose electromechanical systems that used punched cards to input and store data that instructed the machine what to do. These tabulation machines were essentially calculators that counted and summarized information. They did this task well, but only this task.

These machines supported the scaling of business and society and were used in government applications, such as processing population census data, and business applications, such as accounting and inventory control. Tabulating machines evolved to a class of machines that were known as unit record equipment, which were instrumental in starting the data processing industry.

Programming era (1950s – present)

This era started with the shift from mechanical tabulators to electronic systems. It began during World War II and was driven by military and scientific needs. Following the war, digital

“computers” evolved rapidly and moved into businesses and governments. The programmable computing era began.

The significant change of this era is the introduction of general-purpose computing systems that are programmable: They can be reprogrammed to perform different tasks and solve multiple problems in business and society. However, they must be programmed and are still constrained in their interaction with human beings. Everything we now know as a computing device, such as the mainframe, personal computer, smartphone, and the tablet, is a programmable computer. Some experts believe that this era of computing will continue to exist indefinitely.

AI and cognitive era (2011 – future)

AI systems are a natural evolution of programmable computing. AI systems exhibit characteristics of human intelligence. They can understand and reason and learn and improve their performance overtime.

Dr. Kelly in the white paper *Computing, cognition and the future of knowing* defines cognitive computing as follows:

“Cognitive computing refers to systems that learn at scale, reason with purpose and interact with humans naturally. Rather than being explicitly programmed, they learn and reason from their interactions with us and from their experiences with their environment.”

ARTIFICIAL INTELLIGENCE TYPES AND APPROACHES

Most smart software programs can be considered as AI systems because they perform tasks that can be considered intelligent human tasks. Researchers and developers are adopting one of two approaches of thinking while building AI systems:

- Strong AI
- Weak AI

The first approach builds systems that can act and think intelligently like people do. This approach simulates human reasoning and cognition in general tasks and is called *strong AI*. Most of these projects are in the small-to-medium size range.

The three largest projects are:

- DeepMind,

- The Human Brain Project (an academic project that is based in Lausanne, Switzerland), and
- OpenAI.

The second approach is not concerned about whether the AI systems display human-like cognitive functions; the focus is on AI systems that perform specific tasks accurately and correctly. This approach is called *weak AI*. An example of weak AI is a chatbot, which is trained to answer a specific set of questions about a domain. A chatbot does not have self-awareness or genuine intelligence.

Generally, current AI is somewhere between strong and *weak AI*. The current most advanced AI research uses human reasoning as a model but the AI field has not modeled human reasoning yet. A good example of current AI is advanced natural language generation.

EVOLUTION OF ARTIFICIAL INTELLIGENCE TYPES

In terms of scope, Ray Kurzweil, who is a futurist that is working for Google, identifies three levels of AI:

- Artificial Narrow Intelligence (ANI)
- Artificial General Intelligence (AGI)
- Artificial Super Intelligence (ASI)

Artificial Narrow Intelligence (ANI)

ANI, which belongs to the weak AI school, refers to applying AI techniques to narrow problems. It focuses on a specific task. The strength of ANI is that it focuses on doing something extremely well, sometimes exceeding a human's capabilities. ANI is a good fit for automating simple and repetitive tasks.

Examples of ANI are bots and virtual assistants, such as Siri, Microsoft Cortana, Amazon book recommendations that are customized for each user, restaurant recommendations, weather updates, Watson DeepQA, and customer services chatbots for answering simple and repetitive customer inquiries.

These AI implementations excel in their defined areas, but if you try to start a conversation with a virtual assistant, you clearly see how "narrow" its intelligence is.

Artificial General Intelligence (AGI)

AGI belongs to the strong AI school. It refers to computer systems that exhibit capabilities of the human brain. AGI refers to systems or machines that can generally perform any intellectual task that a human can do.

Artificial Super Intelligence (ASI)

ASI refers to machines that surpass humans in general intelligence. Nick Bostrom, a leading AI thinker, in the paper How Long Before Superintelligence?, defines ASI as “an intellect that is much smarter than the best human brains in practically every field, including scientific creativity, general wisdom and social skills.”

Regarding ASI, many prominent scientists and technologists have ethical concerns about the future of humanity and intelligent life. The unique capabilities of the human brain are the reason why humans have a dominant position over other species. Super-intelligent machines might surpass the human brain in general intelligence.

Applications of AI

AI has been dominant in various fields such as –

Gaming – AI plays crucial role in strategic games such as chess, poker, tic-tac-toe, etc., where machine can think of large number of possible positions based on heuristic knowledge.

Natural Language Processing – It is possible to interact with the computer that understands natural language spoken by humans.

Expert Systems – There are some applications which integrate machine, software, and special information to impart reasoning and advising. They provide explanation and advice to the users.

Vision Systems – These systems understand, interpret, and comprehend visual input on the computer. For example,

- o A spying airplane takes photographs, which are used to figure out spatial information or map of the areas.
- o Doctors use clinical expert system to diagnose the patient.
- o Police use computer software that can recognize the face of criminal with the stored portrait made by forensic artist.

Speech Recognition – Some intelligent systems are capable of hearing and comprehending the language in terms of sentences and their meanings while a human talks to it. It can handle different accents, slang words, noise in the background, change in human's noise due to cold, etc.

Handwriting Recognition –The handwriting recognition software reads the text written on paper by a pen or on screen by a stylus. It can recognize the shapes of the letters and convert it into editable text.

Intelligent Robots – Robots are able to perform the tasks given by a human. They have sensors to detect physical data from the real world such as light, heat, temperature, movement, sound, bump, and pressure. They have efficient processors, multiple sensors and huge memory, to exhibit intelligence. In addition, they are capable of learning from their mistakes and they can adapt to the new environment.